

assignment10 cosmic collision

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:32pm

Quiz Instructions



Question 1 6 pts

In the movie "asteroids we worry about" (u-tube channel veritasium), the scientists estimates that the probability for a 10km meteorite (like the one that wiped out the dinosaurs) to hit the Earth during our life time is

☐

1 in a million

☐

1 in 1000

☐

1 in 10

☐

0



Question 2 5 pts

A meteorite could be a comet or an asteroid

☐

False

☐

True



Question 3 5 pts

The "greeks" and the "troyans " share the same the same orbit as

☐

Jupiter

☐

Mars

☐

Kuiper belt

☐

Earth



Question 4 5 pts

A subsurface crater was found in Mexico. It was detected because of its low g_____ values. It was filled with sediments (not as heavy as rocks)

☐

No answer text provided.

☐

magnetic field

☐

electric field

☐

gravity

☐

Question 5 5 pts

The Barringer crater was formed by a _____ size meteorite

☐

50m

☐

10km

☐

100m

☐

1km

☐

Question 6 5 pts

Which of the following statements about comets and asteroids is *true*?

☐

All asteroids lie in the asteroid belt between Mars and Jupiter.

☐

Comets are balls of ice and dust.

☐

Most of the trillions of comets in our solar system have tails.

☐

Only asteroids collide with Earth.

☐

Question 7 3.5 pts

Charon is

☐

a comet

☐

a dwarf planet

☐

an asteroid



a Moon



Question 8 3.5 pts

We can find the mass of an asteroid by finding the _____ of its Moon



size



color



orbital period



mass



Question 9 3.5 pts

Kinetic energy depends on the _____ squared

(going from 30mph to 60 mph, potential damage is x 4)



speed



mass



density



height



Question 10 3.5 pts

The meteorite that killed the dinosaurs was about _____ in size



1km



10km



50m



100m



Question 11 3.5 pts

What is a *meteorite*?

☐

a comet that burns up in Earth's atmosphere

☐

a streak of light caused by a star moving across the sky

☐

a fragment of rock that has fallen to Earth's surface from space

☐

a small moon that orbits one of the jovian planets

☐

a streak of light caused by a small particle from space burning up in Earth's atmosphere



Question 12 3.5 pts

The comet Shoemaker-Levy hit

☐

Jupiter

☐

the Moon

☐

Ceres

☐

The Earth



Question 13 5 pts

Ref.: movie "asteroids we worry about". In 2013, a meteorite landed in a town in Russia. The damages done by the meteorite (injured persons, collapsing building) was due to

☐

sound shock wave

☐

The burst equivalent to an atomic bomb

☐

The crater that formed in the middle of the town.

☐

intense heat from the kinetic energy of the meteorite



Question 14 3.5 pts

In 2006, Ceres was promoted and is called a

☐

Moon

☐

planet

☐

centaur



dwarf planet



Question 15 3.5 pts

Luis Alvarez and team found of thin layer of _____ - worldwide (proof of a huge meteorite hitting the EArth)



uranium



sediments



iridium



iron



Question 16 3.5 pts

The meteorite that killed the dinosaurs delivered about



1 million megatons of TNT



100 million megatons of TNT



25ktons of TNT



1 megaton of TNT



Question 17 7 pts

A crater of the correct age to be from the impact thought to have caused the extinction of the dinosaurs was found in



Northern Africa.



Arizona.



Chelyabinsk, Russia.



Tunguska, Siberia.



the Yucatán Peninsula in Mexico.



Question 18 10 pts

How can a large asteroid impact in one spot on Earth affect the climate of the entire Earth?



The impact tears a hole in Earth's atmosphere, allowing heat energy to escape and thereby cooling our planet.



The impact blasts debris into Earth's atmosphere that blocks sunlight, making Earth's surface cooler.



The impact releases greenhouse gases that immediately make Earth warmer.



The impact blasts a hole through Earth's mantle, allowing hot metal from the core to rise up and heat the world.



Question 19 5 pts

On average, how often do impactors about 10 km in size, large enough to produce mass extinction, hit Earth?



once every century



once every thousand years



once every hundred million years



once in Earth's history



once every million years



Question 20 3.5 pts

The centaurs orbit between



Neptune, Kuiper Belt



Mars, Jupiter



Jupiter, Saturn



Jupiter, Neptune



Question 21 3.5 pts

Why was the Shoemaker-Levy 9 impact on Jupiter so important to astronomers?



It wiped out the dinosaurs.



It was the first event in modern history that was brighter than the full moon in the sky.



It confirmed our theory of solar system formation.



It dredged up material that allowed us to learn about Jupiter's interior.



It was the first clear evidence that impacts really occur.



Question 22 5 pts

Toutatis, the asteroid, was name after



the brighther star in the sky



The goddess of war



A joyful centaur



a celtic god



Question 23 5 pts

The meteorite that caused the extincion of the dinosaurs also caused the extinction of other species



True



False



dinosaurs?? what's that?



we don't know

Not saved

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